



CURSO: CALCULO INTEGRAL

COD. CURSO: BMA-02MN

TIPO DE PRUEBA: PRACTICA No. Ex. PARCIAL Ex. FINAL Ex. SUST.

Calcule las siguientes integrales

1. $\int \frac{e^x(1-5x+x^2+\text{sen}(x)-5x\cos(x)+x^2\cos(x))}{1+\cos(x)} dx \dots (3 \text{ p})$

2. $\int \frac{3\text{sen}(x)\sec^2(x)-4\text{sen}(x)\sec(x)-4\text{sen}(x)}{(\cos(x))^{-1}-1-4\cos(x)+4\cos^2(x)} dx \dots (3 \text{ p})$

3. $\int \frac{[\arcsen(x)\ln(\arcsen(x))]^2}{\sqrt{1-x^2}} dx \dots (2 \text{ p})$

4. $\int \frac{\tanh(x)dx}{(\cosh(x)-1)^3(\cosh(x)+2)^4} \dots (3 \text{ p})$

5. $\int \frac{e^{15x}dx}{(e^{5x}+3)^3} \dots (3 \text{ p})$

6. $\int \frac{(x^2+3x-2)dx}{(x^2-6x+13)^3} \dots (3 \text{ p})$

7. $\int \frac{\cos(x)dx}{1-3\text{sen}(x)+2\cos(x)} \dots (3 \text{ p})$

LOS PROFESORES DEL CURSO